

Gibbs NWEMADJI TIAKO

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Research Interest

Machine Learning, Statistical Inference, Deep Learning, Data Science, Information Theory, Statistical Physics, Quantum Information, Quantum Machine Learning.

Preprints and Publications

When pre-training hurts LoRA fine-tuning: a dynamical analysis via single-index models Feb 2026

Gibbs Nwemadji, Bruno Loureiro, Jean Barbier. arXiv:2602.02855

Generalization performance of narrow one-hidden layer networks in the teacher-student setting Apr 2026

Rodrigo Pérez Ortiz, Gibbs Nwemadji, Jean Barbier, Federica Gerace, Alessandro Ingrosso, Clarissa Lauditi, Enrico M. Malatesta. Journal of Statistical Mechanics: Theory and Experiment

On the Capacity of a Quantum Perceptron for storing Biased Patterns Dec 2024

Fabio Benatti, Giovanni Gramegna, Stefano Mancini and Gibbs Nwemadji, Journal of Physics A.

Research Visits

Yale University, New Haven, CT, USA

Mar 2026–Jul 2026

Visiting PhD Student

I am currently visiting Prof. John Sous to study how superposition induces scaling laws in deep neural networks, and we also aim to understand the representations induced by LoRA fine-tuning in LLM.

École Normale Supérieure (ENS–PSL), Paris, France

Mar 2025–May 2025

Visiting PhD Student

I worked with Prof. Bruno Loureiro on developing a theory for Low-Rank Adaptation fine-tuning, for which the first preprint is now available, and we are currently extending this work to more complex settings.

Education

Scuola Internazionale Superiore di Studi Avanzati (SISSA), Italy. Oct 2022–Dec 2026

PhD. in Theoretical and Scientific Data Sciences

Thesis Title: Topic for Low-Rank Adaptation and Neural Networks

Supervisor: Prof. Jean Barbier

Relevant Courses:

○ Unsupervised Learning and non-Parametric methods (A+) ○ Theory of Neural Networks (A)

The Abdus Salam International Centre for Theoretical Physics (ICTP), Italy. Sep 2021–Aug 2022

Postgraduate Diploma Program in Physics (Major: Quantitative Life Sciences) (Valedictorian)

Thesis Title: Storage Capacity of a Quantum Perceptron with Biased Patterns (A+)

Supervisor: Prof. Fabio Benatti

African Institute for Mathematical Sciences (AIMS), Cameroon. Oct 2020–Jul 2021

STRUCTURED MASTER'S DEGREE in Mathematical Sciences (Major: Data Science)

Thesis Title: Non-Boolean Quantum Amplitude Amplification Algorithm (A+)

Supervisor: Dr. Wolfgang Scherer

University of Douala (UD), Cameroon.

Oct 2015–Sept 2020

MASTER'S DEGREE in Physics (Major: Science and Radiation Matter)

BACHELOR'S DEGREE in Physics (Major: Fundamental Physics)

Research Achievements (Honours/Awards)

SISSA (International School for Advanced Studies)

Oct 2022–2026

- Recipient of a fully funded Ph.D. scholarship (tuition and stipend) to pursue doctoral studies at SISSA, Trieste, Italy.

ICTP/UNESCO

Sep 2021–Aug 2022

- Best student of the Abdus Salam ICTP Diploma (Pre-PhD) program in Quantitative Life Science.
- Recipient of the fully funded intensive Pre-PhD program at the Abdus Salam ICTP, Trieste.

Next Einstein Initiative

Oct 2020–Jul 2021

- Recipient of the fully funded Master in Data Sciences at the African Institute for Mathematical Sciences.

University of Douala

Oct 2015– Jun 2019

- Best student of the year prize awarded by the Chancellor of the University of Douala.
- Awarded four times with the Excellence Prize of the President of the Republic of Cameroon.

School and Training (Poster and Participation)

- ROCKIN' AI 2025 1-5 Sept 2025
- Youth in High-Dimensions: Recent Progress in Machine Learning, High-Dimensional Statistics and Inference & How creative is Generative AI? Perspectives from Science and Philosophy, (ICTP, smr 4084) 3-10 Jul 2025
- Towards a Theory for Typical-Case Algorithmic Hardness: Les Houches Jan 27 - Feb 7, 2025
- ROCKIN' AI 2024 2-6 Sept 2024
- Youth in High-Dimensions: Recent Progress in Machine Learning, High-Dimensional Statistics and Inference, (ICTP, smr 3841) 20-24 May 2024
- Workshop on Complex Systems, Statistical Mechanics and Machine Learning Crossover:les Houches 24-29 March 2024
- Workshop on Learning and Inference from Structured Data: Universality, Correlations and Beyond, (ICTP, smr 3850) 3 Jul - 7 Jul 2023
- Youth in High-Dimensions: Recent Progress in Machine Learning, High-Dimensional Statistics and Inference, (ICTP, smr 3841) 21 May - 2 Jun 2023
- Workshop on Ethical and Societal Challenge of Machine Learning (ICTP, smr 3749) 7 Nov - 11 Nov 2022
- Spring College on Physics of Complex Systems (SISSA-ICTP: SMR 3690) 21Feb - 15 Mar 2022

Professional/Teaching Experience

High School Teacher(Adventist Bilingual High School Douala Logpom)

Sept 2019 – Aug 2020

- I have taught mathematics, physics, and chemistry to high school students.

Achievements/Certificates

IBM Quantum Challenge - Africa 2021

Sept 2021

QBRONZE by QWorld (Quantum Computing and Programming using Qiskit)

Jun 2021

Zindi Africa Hackathon

Jan 2021- Apr 2021

- AIMS Data Science Hackathon Rank:10/123.

Python (IBM certification)

Nov 2020

Skills and Competences

Programming

- Python(Pytorch, keras,...), R, Fortran, Math-Lab, Latex.

Language

- French: Native, English: C1 Italian: B1.

Software

- SageMath, Rstudio, MS Office, TexStudio.

Hobby

- Book, Basketball, Run.

Extracurricular

OneQuantum Cameroon: President

Sept 2021 – 2022

- Make knowledge of quantum science available everywhere in Cameroon.

IAPS(International Association of Physics Student) Cameroon: Vice-President

April 2021 – 2022

- Making physics known throughout the world and in Cameroon.
- Solving real-world problems with physics.

Library Club: Co-President (AIMS-Cameroon)

Oct 2020 – Jul 2021

- Educating children and illiterates about efficient reading methods..

Peer Tutor

Sept 2014 – Present

- Undergraduate students and I meet weekly to discuss courses related to mathematics, physics and computer science.
- Tutoring high school students in math and science.